

Spooky Problem Set

UMass Recreational Math Club

September 2025

Problem 1. Last night, somebody broke into Lederle Graduate Research Tower and axe murdered everyone inside. The police currently have three suspects: Sam, Nat, and Ross. They made the following statements:

- Sam: I'm innocent.
- Nat: I'm innocent.
- Ross: Nat is guilty.

Only one of these statements is true. Who is innocent?



Problem 2. The Grim Reaper knocks on a woman's door, and sees her and her three children. He asks for the ages of her children. She tells him, "The product of their ages is 72. The sum of their ages is the number on the door." So the Grim Reaper goes outside and checks the number of the door. He comes back and says, "I need more information." She replies: "The oldest one likes strawberries." Immediately, the Grim Reaper figures out their ages. What are their ages?



Problem 3. On a dark and spooky night three students break into LGRT to steal some Hagoromo chalk. The first student steals half of the pieces of chalk in the box, plus half a piece of chalk. The second student steals half of the remaining chalk plus a half piece of chalk. The final student took what was left: 1 piece of chalk. How many pieces of chalk were there initially?



Problem 4. There are one hundred MATH 300 students standing in a circle, with one next to two, etc, and one hundred next to one. They count off beginning at one and ending at one hundred. One has a sword and kills two. She passes the sword to three who kills four, and so forth. Ninety-nine kills one hundred and passes the sword to one. Then one kills three and passes the sword to five. This goes on until only one person is left standing. Which number are they?